

1. In relational databases a transaction is...
 - A set of operations that forms an inseparable unit.
2. What is an implicit transaction?
 - A transaction that is managed by DBMS and requires no explicit commands for start, commit or roll back.
3. Do all DBMSes support implicit transactions?
 - Yes, except for Oracle.
4. How can we detect if implicit transactions are enabled?
 - Each DBMS has its own non-trivial mechanisms
5. Can we still manipulate database data with implicit transactions disabled?
 - Yes.
6. Do all DBMSes support explicit transactions?
 - Yes.
7. Can we commit an explicit transaction?
 - Yes
8. Can we roll back an explicit transaction?
 - Yes
9. How can we detect if explicit transactions are enabled?
 - We don't need to detect if explicit transactions are enabled.
10. Can we manipulate database data with explicit transactions?
 - Yes
11. A transaction isolation level is...
 - A virtual value showing the degree of visibility of changes between transactions
12. The "read committed" transaction isolation level protects us from the following problems:
 - Lost update
 - Dirty read.
13. The "repeatable read" transaction isolation level protects us from the following problems:
 - Lost update.
 - Dirty read
 - Non-repeatable read
14. The "read uncommitted" transaction isolation level protects us from the following problem
 - Lost update

15. The “snapshot” transaction isolation level protects us from the following problems:
 - Lost update.
 - Dirty read.
 - Non-repeatable read
16. So called “lost update” problem arises when...
 - One transaction loses its update due to other’s transaction later update
17. So called “dirty read” problem arises when...
 - One transaction reads temporary data produced by another transaction (such data will never be saved in the database)
18. So called “non-repeatable read” problem arises when...
 - A transaction gets different results on the first and the second read attempts
19. So called “phantom reads” problem arises when...
 - A transaction gets different results of some data aggregation on the first and the second read attempts
20. Can a DBMS handle mutual transaction lock situation?
 - Yes, one of such transactions will be rolled back
21. Can we commit or roll back a transaction from a stored function?
 - No, but we may create a situation that forced a DBMS to roll back a transaction
22. Can we commit or roll back a transaction from a stored procedure?
 - Yes
23. Can we commit or roll back a transaction from a trigger?
 - Not in every DBMS, but we may create a situation that forced a DBMS to roll back a transaction
24. Can we change a transaction isolation level in a stored procedure?
 - Yes
25. Can we change a transaction isolation level in a stored function?
 - No